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HW 12

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#1) Laminar Flow: $\frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} = ?$

$$\bullet \langle V_z \rangle = \frac{\int dQ}{A} = \int_0^R \frac{V_z 2\pi r dr}{\pi R^2} = \int_0^R \frac{V_{z,max} (1 - \frac{r^2}{R^2}) 2\pi r dr}{\pi R^2}$$

$$\Rightarrow \frac{2}{R^2} \int_0^R (1 - \frac{r^2}{R^2}) r dr = \frac{1}{2} r^2 - \frac{1}{4} \frac{r^4}{R^2} \Big|_0^R = \frac{1}{2} R^2 - \frac{1}{4} \frac{R^4}{R^2}$$

$$\bullet \langle V_z \rangle = \frac{2}{R^2} V_{z,max} (R^2 - \frac{1}{4} R^2) = \frac{1}{2} V_{z,max} \quad (1)$$

$$\langle V_z \rangle^2 = \left[\frac{1}{2} V_{z,max} \right]^2 = \frac{1}{4} V_{z,max}^2 \quad (2)$$

$$\text{or } V_z = V_{z,max} (1 - \frac{r^2}{R^2})$$

$$\langle V_z^2 \rangle = \frac{\int dQ}{A} = \int_0^R \frac{V_z^2 2\pi r}{\pi R^2} dr = \int_0^R \frac{(V_{z,max} [1 - \frac{r^2}{R^2}])^2 2\pi r}{\pi R^2} dr$$

$$\Rightarrow (V_{z,max} [1 - \frac{r^2}{R^2}])^2 = V_{z,max}^2 - 2V_{z,max}^2 \frac{r^2}{R^2} + V_{z,max}^2 \frac{r^4}{R^4}$$

$$\Rightarrow \frac{V_{z,max}^2 2\pi}{\pi R^2} \int_0^R (r - \frac{2r^3}{R^2} + \frac{r^5}{R^4}) dr = \frac{V_{z,max}^2 2\pi}{\pi R^2} \left(\frac{1}{2} r^2 - \frac{1}{2} \frac{r^4}{R^2} + \frac{1}{6} \frac{r^6}{R^4} \right) \Big|_0^R$$

$$\Rightarrow \frac{V_{z,max}^2 2\pi}{\pi R^2} \left(\frac{1}{2} R^2 - \frac{1}{2} \frac{R^4}{R^2} + \frac{1}{6} \frac{R^6}{R^4} \right) = V_{z,max}^2 \left(1 - 1 + \frac{1}{3} \right)$$

$$= \frac{1}{3} V_{z,max}^2 \quad (3)$$

$$\bullet \frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} = \frac{\frac{1}{3} V_{z,max}^2}{\frac{1}{4} V_{z,max}^2} = \boxed{\frac{4}{3}}$$

Turbulent Flow: $V_z = V_{z,max} (1 - \frac{r}{R})^{1/7}$

$$\bullet \langle V_z^2 \rangle = \frac{\int dQ}{A} = \int_0^R \frac{V_z^2 2\pi r}{\pi R^2} dr = \int_0^R \frac{(V_{z,max} (1 - \frac{r}{R})^{1/7})^2 2\pi r}{\pi R^2} dr$$

$$\Rightarrow \frac{V_{z,max}^2 2\pi}{\pi R^2} \int_0^R (1 - \frac{r}{R})^{2/7} r dr = \frac{V_{z,max}^2 2\pi}{\pi R^2} \left(\frac{7(R-r)^{9/7} (9r + 7R)}{144 r^{2/7}} \right) \Big|_0^R$$

$$\Rightarrow \frac{V_{z,max}^2 2\pi}{\pi R^2} \left(\frac{49R^2}{144} \right) = \frac{V_{z,max}^2 49}{72} \quad (1)$$

$$\bullet \langle V_z \rangle = \frac{\int dQ}{A} = \int_0^R \frac{V_{z,max} 2\pi r}{\pi R^2} dr = \int_0^R \frac{V_{z,max} (1 - \frac{r}{R})^{1/7} 2\pi r}{\pi R^2} dr$$

$$\Rightarrow \frac{V_{z,max} 2\pi}{\pi R^2} \int_0^R (1 - \frac{r}{R})^{1/7} r dr = \frac{V_{z,max} 2\pi}{\pi R^2} \left(\frac{49R^2}{180} \right)$$

$$= \frac{V_{z,max} 49}{60} \quad (2)$$

$$\bullet \frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} = \frac{\left[\frac{44 V_{z, \max}^2}{72} \right]}{\left[\frac{44 V_{z, \max}^2}{60} \right]^2} = \frac{50}{49} \checkmark$$

$$\#2) \frac{\bar{V}_z}{V_{z, \max}} = \left(1 - \frac{r}{R}\right)^{\frac{1}{n}}$$

we have to prove:

$$\frac{\langle \bar{V}_z \rangle}{V_{z, \max}} = \frac{2n^2}{(n+1)(2n+1)}$$

$$\begin{aligned} \bullet \bar{V}_z &= \frac{2V_{z, \max}}{R^2} \int_0^R \left(1 - \frac{r}{R}\right)^{\frac{1}{n}} r dr = \frac{n(R-r)^{\frac{1}{n}}(r-R)(nr+r+nR)}{(n+1)(2n+1)R^{\frac{1}{n}}} \Big|_0^R \\ &= \frac{(0) - n(R)^{\frac{1}{n}}R(nR)}{(n+1)(2n+1)R^{\frac{1}{n}}} \end{aligned}$$

$$\bullet \frac{2V_{z, \max}}{R^2} = \frac{n^2 R^{\frac{1}{n}+2} R^2}{(n+1)(2n+1) R^{\frac{1}{n}}}$$

$$\langle \bar{V}_z \rangle = 2V_{z, \max} \frac{n^2}{(n+1)(2n+1)} \checkmark$$

$$\bullet \frac{\langle \bar{V}_z \rangle}{V_{z, \max}} = \frac{2n^2}{(n+1)(2n+1)} \checkmark$$